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MILITARY SPECIFICATION

REEL, TOW-TARGET, 12,000-FOOT CABLE CAPACITY, AIRBORNE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification establishes the requirements for one type of airborne tow-target cable reel, as shown on Drawing AN8030, capable of reeling in, storing, holding and reeling out 12,000 feet of 1/8-inch-diameter steel cable used in target towing operations.

2. APPLICABLE SPECIFICATIONS, OTHER PUBLICATIONS, AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Military

MIL-D-5028	Drawings and Data Lists; Preparation of (for Engines, Accessories, and Other Auxiliary Equipment)
JAN-P-105	Packaging and Packing for Overseas Shipment - Boxes; Wood, Cleated, Plywood
JAN-P-106	Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed
JAN-P-125	Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible
JAN-P-139	Packaging and Packing for Overseas Shipment - Plywood, Container-Grade
JAN-P-140	Packaging and Packing for Overseas Shipment - Adhesives; Water-Resistant, Case Liner

U. S. Army

94-40645	Marking; Exterior, Domestic and Export Shipment, by Contractors

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2.1.2 Other Publications.-

Air Force-Navy Aeronautical Bulletin

No. 143 Specifications and Standards - Use of

Bureau of Supplies and Accounts

Navy Shipment Marking Handbook

2.1.3 Drawings.-

Air Force-Navy Aeronautical Standard Drawings

AN8030 Reel - Tow-Target, 12,000 Foot Cable Capacity, Airborne

Air Force-Navy Aeronautical Standard Issue of BuAer Drawing

No. 388687 Reel; Tow-Target, 12,000-Foot Cable Capacity, Airborne - General Arrangement for

(Copies of this specification and copies of other publications referenced herein or required for Government procurement, and the Index of Military Aeronautical (AN or MIL) Standards, may be obtained upon application to the Commanding General, Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio; or the Commanding Officer, U. S. Naval Air Station, Johnsville, Pennsylvania.)

3. REQUIREMENTS

3.1 Materials.- Materials shall conform to applicable specifications as specified herein. Materials which are not covered by applicable specifications or which are not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.1.1 Metals.- All metals used in the construction of reels shall be of a corrosion-resisting type or shall be suitably protected to resist corrosion during the normal service life of the reels. The use of dissimilar metals, especially brass, copper, or steel, in contact with aluminum or aluminum alloy shall be avoided whenever practicable. When such contacts are unavoidable, the metal shall be properly protected against electrolytic corrosion.

3.1.2 Selection of Materials.- Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.1.2.1 AN or JAN Standard Parts.- AN or JAN Standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable properties and are replaceable by the AN or JAN Standard parts without alteration, and provided the corresponding AN or JAN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding AN or JAN Standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

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3.2 Design and Construction.-

3.2.1 General.- Airborne tow-target cable reels shall, in general, consist of a frame, drum, chain-drive assembly, fluid motor, clutch, brake, and level wind assembly. Assembly and all detail parts of reels shall be in accordance with all the requirements contained in the applicable Air Force-Navy Aeronautical Standard Issue of the Bureau of Aeronautics Drawing specified in Drawing AN8030. The reel shall be constructed to withstand, without failure, abuse such as dropping of hand tools, and personnel standing on the reel frame, and such other conditions as are incident to shipping, stowage, installation, and service.

3.2.2 Finish.- Completed reels shall be free from sharp edges and burrs and shall be finished in accordance with the applicable drawings and specifications.

3.2.3 Assembly.- Reels shall be completely assembled without interference, binding, or misalinement of any parts.

3.3 Interchangeability.- All parts having the same part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.4 Operating Conditions.- Reels shall be capable of satisfactory operation under any combination of the following conditions when manufactured in accordance with the applicable drawings and specifications:

(a) Temperatures: Under ambient temperatures of between -54°C (-65°F) and 71°C (160°F).

(b) Mounting: With reel drum axis tilted 30 degrees from the horizontal.

(c) Pressure: Under atmospheric pressures prevailing at altitudes between sea level and 40,000 feet above sea level.

(d) Acceleration: At accelerations up to 4 positive gs and 2 negative gs.

(e) Moisture: Under conditions of moisture such as would result if the reel were soaked by rain and operation commenced immediately thereafter.

(f) Vibration: When subjected to vibration attendant to all normal aircraft installations.

3.5 Performance.- Reels shall satisfy the performance requirements specified in Section 4 when subjected to the tests headed as follows:

(a) Capacity
(b) Static Load
(c) Dynamic Load
(d) Life Expectancy

3.6 Identification of Product.- A nameplate, permanently and legibly filled in with the following information, shall be securely attached to the reel frame.

REEL, TOW-TARGET, 12,000-FOOT CABLE CAPACITY, AIRBORNE
AN8030- (proper dash number)
Stock No. (USAF or Navy, as applicable)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

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3.6.1 Detail parts shall be marked as specified on the applicable drawings and in accordance with the requirements of Specification MIL-D-5028.

3.7 Workmanship.- All details of workmanship shall be in accordance with high-grade commercial practice for similar mechanisms.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- All the tests required for the testing of tow-target reels are classified as Inspection tests, for which necessary sampling techniques and methods of testing are specified in this section. The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Procuring Service. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.2 Examination of Product.- Each reel shall be carefully examined to determine conformance with this specification with respect to design and construction, finish, assembly, material, workmanship, and marking.

4.3 Sampling Tests.- The first two production reels of the first contract let to any one contractor and, at the discretion of the Inspector, any reel manufactured under subsequent contracts by the same contractor shall be subjected by the Procuring Service to the following tests in addition to Examination of Product. Representative samples shall be forwarded to the Air Materiel Command, Wright-Patterson Air Force Base, Dayton, Ohio, or to the Aviation Armament Laboratory, Naval Air Development Center, Johnsville, Pennsylvania, whichever is the laboratory of the applicable Procuring Service.

4.3.1 Capacity.- Under any combination of conditions as specified in paragraph titled "Operating Conditions" and with the level wind mechanism operative, the reel shall store evenly not less than 12,000 feet of 1/8-inch-diameter steel towing cable such as is used in target towing operations under a cable tension of 25 pounds.

4.3.2 Static Load.- Under any combination of conditions specified in paragraph titled "Operating Conditions" and with the brake set, reels shall withstand steady or surge cable loads equal to the breaking strength (2,000 pounds minimum) of 1/8-inch-diameter steel towing cable such as is used in target towing operations.

4.3.3 Dynamic Load.-

4.3.3.1 Reeling Out.- Under any combination of conditions as specified in paragraph titled "Operating Conditions," reels shall reel out evenly 12,000 feet of 1/8-inch cable such as is used in target towing operations under a cable load of 400 pounds and limit the lineal velocity of the cable, relative to the reel frame, to a maximum of 4,000 feet per minute. Reels shall also withstand a surge load of 2,000 pounds such as might result if the cable were suddenly stopped by application of the brake at any point during the course of its travel.

4.3.3.2 Reeling In.- Under any combination of conditions as specified in paragraph titled "Operating Conditions," reels shall reel in 12,000 feet of 1/8-inch cable such as is used in target towing operations, and stow the cable evenly on the reel drum under a cable load of 400 pounds and at an average lineal velocity of 700 feet per minute relative to the reel frame. Under cable loads of less than 400 pounds, the cable speed shall increase in proportion to the decrease in cable loading up to a cable speed consistent with the maximum safe speed of the fluid motor specified in the applicable drawings. When driving power is cut off, cable reels shall automatically prevent the cable from reeling out at any point during the reeling in process without application of the brake.

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4.3.4 Life Expectancy.- Reels shall be subjected to 100 operating cycles as outlined in the paragraph entitled "Operating Cycle" without requiring replacement of any parts except the brake lining. Adjustment for clutch and brake wear may be made as required.

4.3.4.1 Operating Cycle.- The following 1-hour cycle is representative of the operating procedure and shall be used for the "Life Expectancy" test:

(a) Six minutes under conditions specified in paragraph titled "Reeling Out."

(b) Thirty-seven minutes under conditions specified in paragraph titled "Static Load."

(c) Seventeen minutes under conditions specified in paragraph titled "Reeling In."

4.3.5 Rejection and Retest.- Any defects that are discovered during the tests of representative samples that are considered due to poor materials or workmanship shall be corrected by the contractor and these corrections incorporated into the construction of all reels covered by the contract at the contractor's expense. Any reel failing to meet the requirements of the individual tests shall be rejected and returned at the contractor's expense. Reels which have been rejected may be replaced or repaired to correct the defects and resubmitted for all specified tests. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the original defects shall be furnished the Inspector. Reels rejected after retest shall not be resubmitted without the specific approval of the Procuring Service. Tests of representative samples shall be completed by the Procuring Service within the guaranty period specified in the contract and the contractor notified as to acceptance or rejection within this period.

5. PREPARATION FOR DELIVERY

5.1 Application.- The preservation, packing, and marking requirements specified herein apply only to direct purchases by or direct shipment to the Government.

5.2 Preservation.- Each reel shall be lubricated with the Service lubricant prescribed by Air Force-Navy Aeronautical Standard Issue of Bureau of Aeronautics Drawing No. 388687, except that oil reservoirs shall be slushed with the operating lubricant and drip-drained prior to replacing the filler cap.

5.3 Packing.- Unless otherwise specified, all reels shall receive domestic packing. Shipping containers shall contain one reel, shall be uniform in size and shall be securely packed.

5.3.1 Domestic Packing.- Unless otherwise specified, each reel shall be packed in a substantial nailed wood or cleated plywood box so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery. The reel shall be mounted on two pieces of 1-1/2 inch OD steel tubing which shall be securely fastened within the box. Except as specified herein, the container shall conform to the requirements of Consolidated Freight Classification Rules in effect at the time of shipment. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.3.2 Overseas Packing.- Unless otherwise specified, each reel shall be packed in a cleated plywood or nailed wood box conforming to Specification JAN-P-105 or JAN-P-106, respectively. The reel shall be mounted on two pieces of 1-1/2 inch OD steel tubing which shall be securely fastened within the box. Plywood, if used, shall conform to Specification JAN-P-139, type A or B, condition I. A case-liner fabricated from barrier material conforming to Specification JAN-P-125 and sealed with adhesive conforming to Specification JAN-P-140 shall be provided. The box shall be strapped to conform to the requirements in Appendix of JAN-P-105 or JAN-P-106, whichever is applicable.

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5.4 Marking and Labeling.-

5.4.1 Shipping Containers.- Each shipping container shall be durably and legibly marked to indicate method of packing (domestic or overseas), shall be marked for shipment in accordance with the requirements applicable to the individual Services, as specified in Specification 94-40645 for the Air Force, and the Navy Shipment Marking Handbook for the Navy and as follows:

REEL, TOW-TARGET, 12,000-FOOT CAPACITY, AIRBORNE
AN8030- (proper dash No.)
Stock No. (USAF or Navy, as applicable)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Name of Manufacturer
Name of Contractor (if different from manufacturer)

6. NOTES

6.1 Intended Use.- Airborne tow-target cable reels are intended for use in military aircraft for reeling in, storing, holding and reeling out 12,000 feet of 1/8-inch cable used in target towing operations.

6.2 Ordering Data.- Requisitions, contracts, and orders should state the AN Part number, whether overseas packing is desired, and should specify the disposition of reels manufactured under contract or order while awaiting completion of the tests of representative samples by the Procuring Service.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Navy-- BuAer

Other interest:
Air Force

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